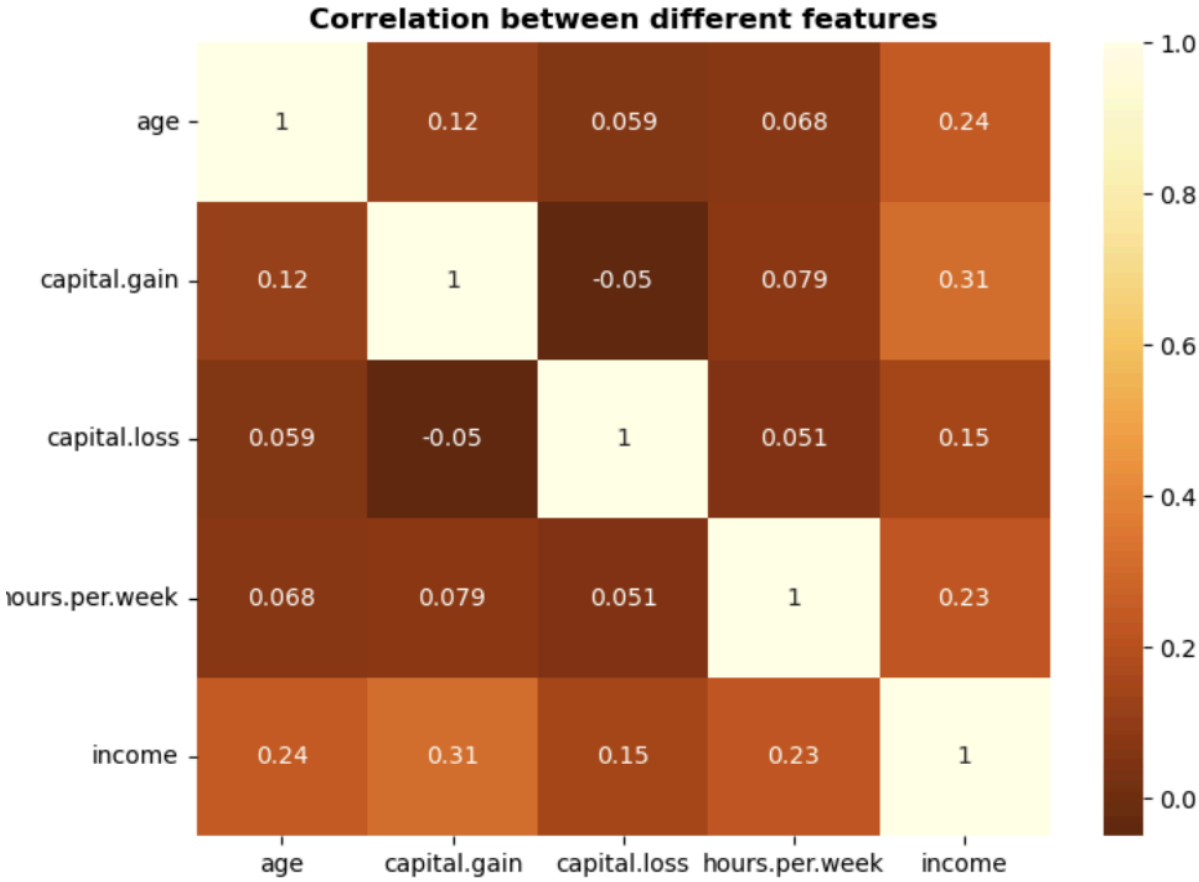
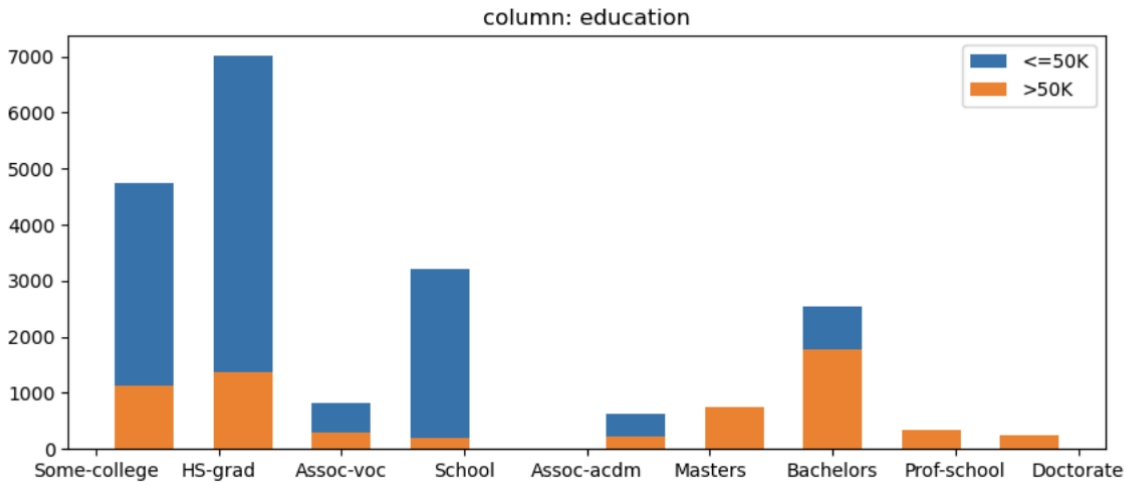
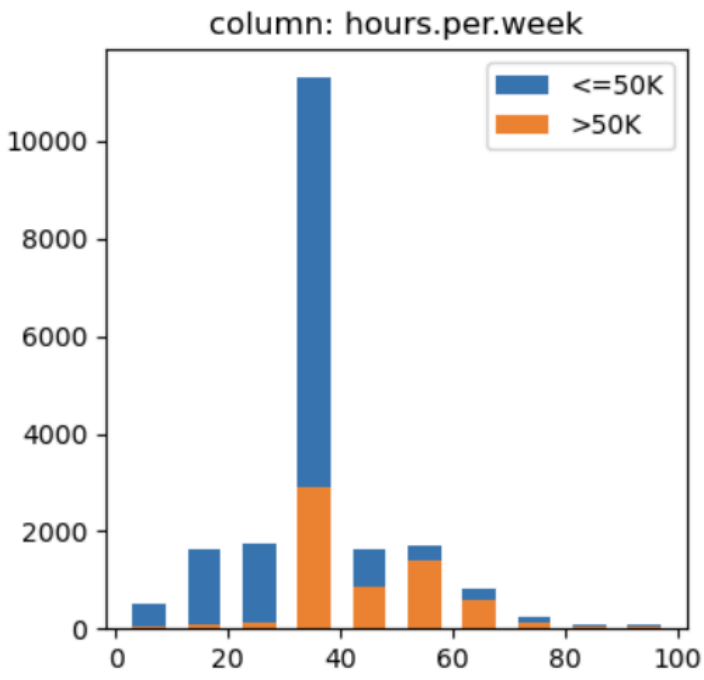
In the heatmap below, it breaks down the correlations for each numerical column and is filtered by color to show which correlations are stronger than the others. We can see that our numerical colors aren't very strongly correlated. For income, it has the strongest correlation with capital gain (0.31), and the weakest correlation with capital loss (0.15).

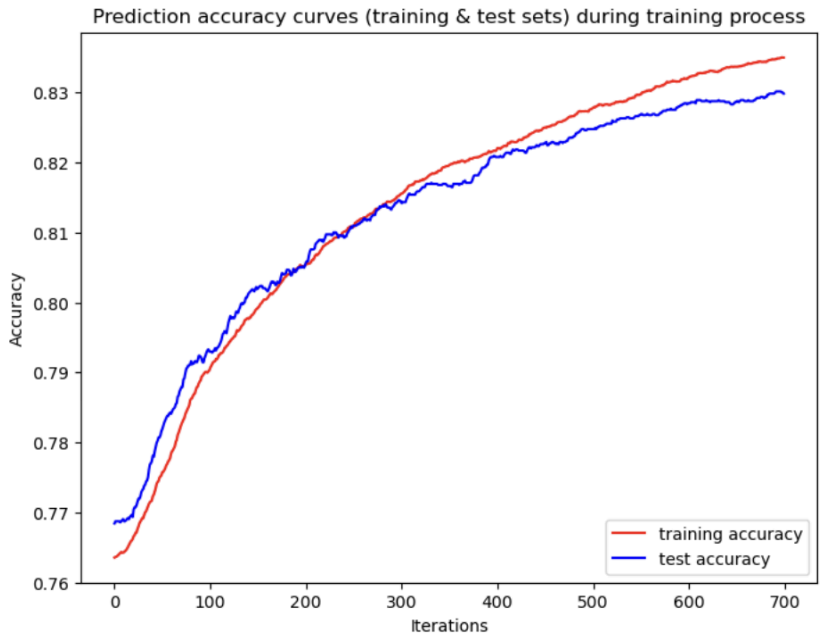


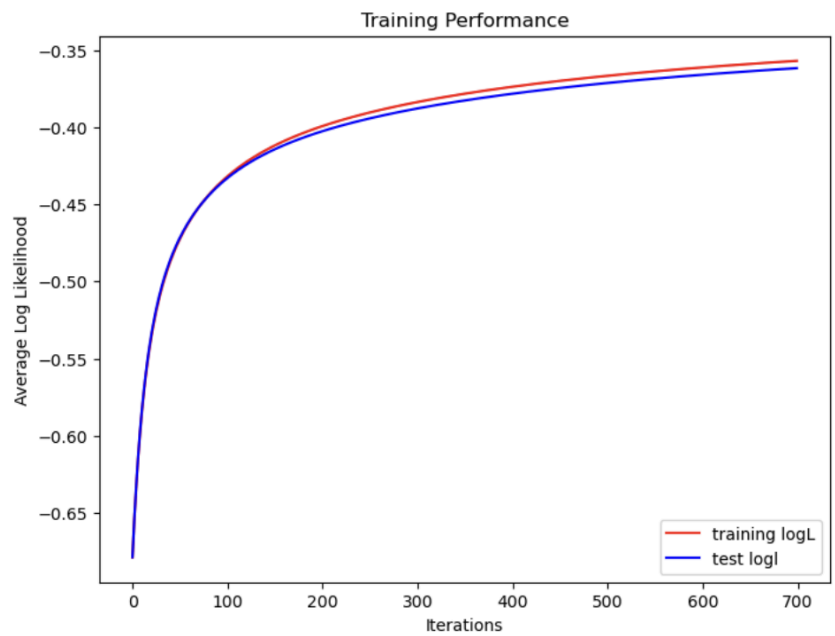
For the Education histogram, we can see that you are more likely to earn more than $50k a year with a higher degree, those being: Masters, Bachelors, Prof-school, and doctorate degree. We can see that a large portion of HS graduates, some college education, and just basic school education earn less than $50k, meaning that with more schooling and higher degree, the more likely you are to get paid more. For the hours per week histogram, we can see that people who work more than 40 hours a week are more likely to earn over $50k a year, while working under 40 hours, you're more likely to earn less than $50k. These histograms both support the fact that if you work more than the average and pursue a higher education, you are more likely to get paid more than $50k a year.



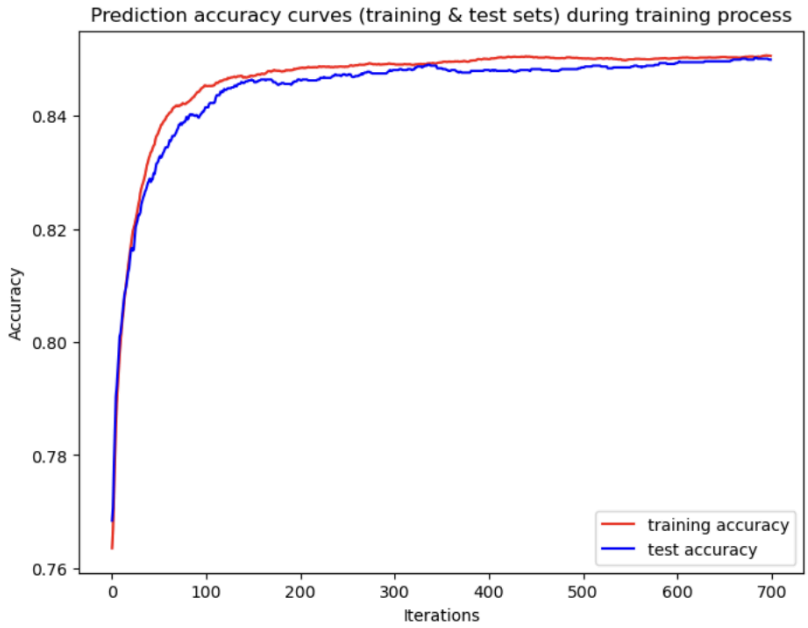


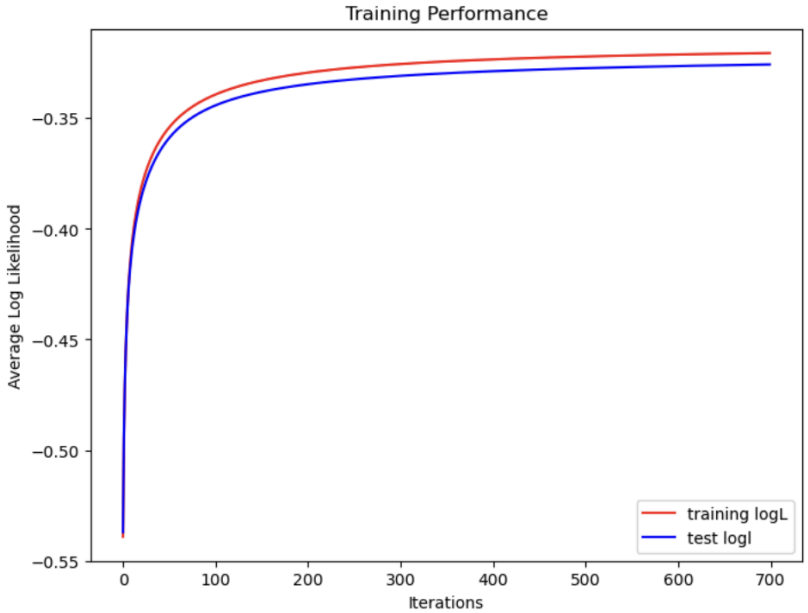
For the learning rate = 0.05, we can see that both the training set and test set follow a similar pattern, increasing gradually over iterations and becoming more accurate. The training set reaches this slightly faster than the test set.

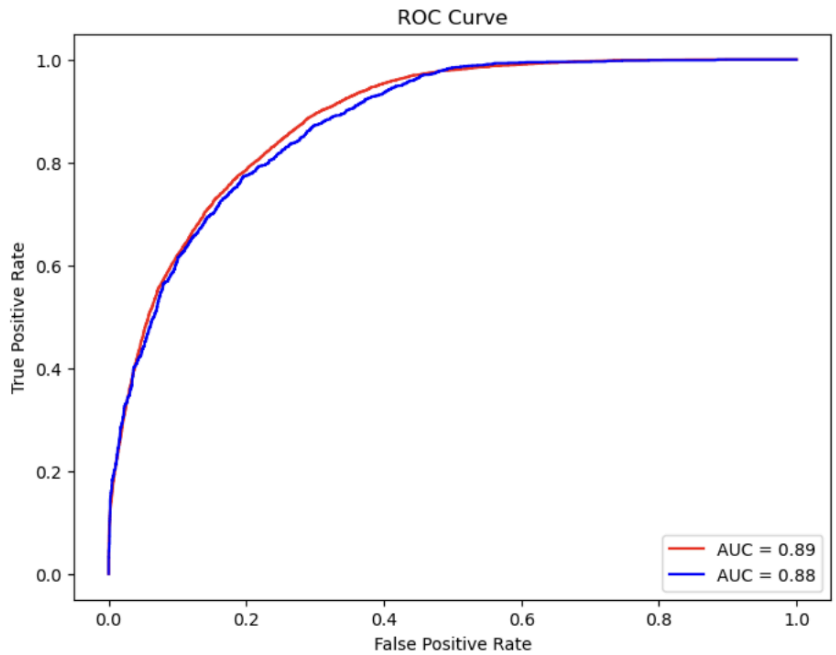
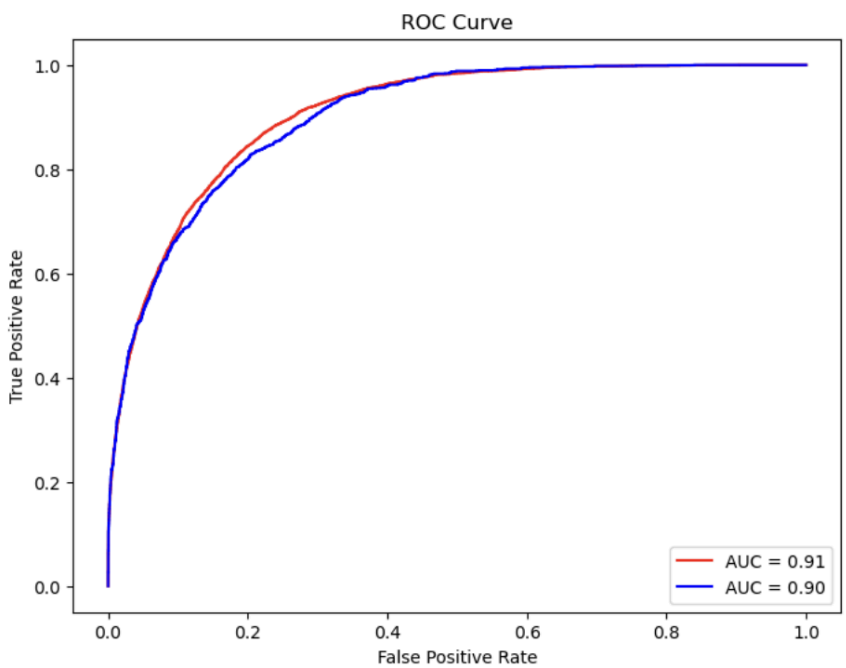




For our learning rate = 0.75, we can see a clearly that our model converges a lot faster than when our learning rate is 0.05. We still see similar patterns where our training set is slightly faster than our test set, but the margin isn't huge. Overall, both learning rates still reach this point, and our training performance is very similar for both.





Top graph: learning rate = 0.05
Bottom graph: learning rate = 0.075

With both of these ROC figures, we can also see (similarly to above), that when our learning rate is greater, we see better results. In this case, we can see that our train and test sets have greater AUC values when our learning rate is 0.75 compared to when it is 0.05 (difference: 0.02). Regardless, we still see on both graphs close alignment and patterns for the train and test sets.